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(54) **GAMING DEVICE HAVING MULTIPLE SPINS FOR WINNING OUTCOMES**

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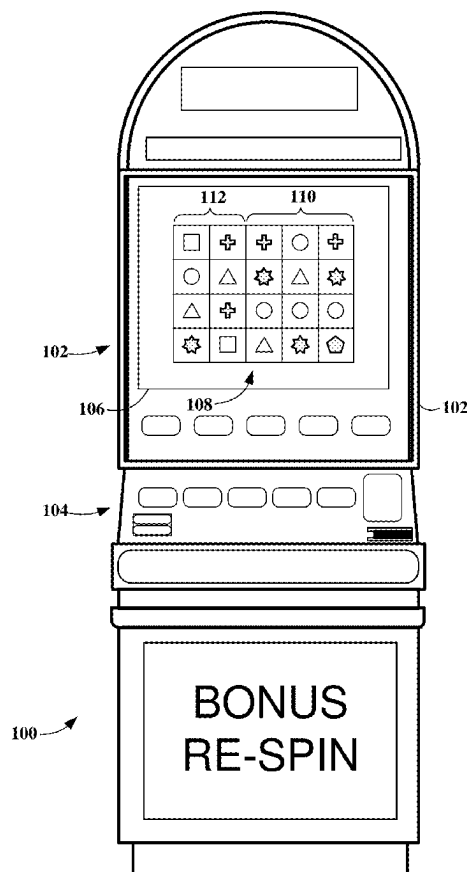
Related U.S. Application Data

(63) Continuation of application No. 18/581,063, filed on Feb. 19, 2024, now Pat. No. 12,307,857, which is a continuation of application No. 15/426,040, filed on Feb. 6, 2017, now Pat. No. 11,908,280, which is a continuation of application No. 14/923,417, filed on Oct. 26, 2015, now Pat. No. 9,564,002, which is a continuation of application No. 13/975,556, filed on Aug. 26, 2013, now Pat. No. 9,171,427, which is a continuation-in-part of application No. 12/077,719, filed on Mar. 20, 2008, now Pat. No. 8,517,813.

(60) Provisional application No. 60/919,362, filed on Mar. 22, 2007.

(57) **ABSTRACT**

Embodiments of the present invention set forth systems, apparatuses and methods for providing multiple spins during gaming events to provide increased opportunities to receive winning outcomes. Accordingly, a gaming device can be configured to have a plurality of reels with multiple symbols on each reel. A portion of the symbols on each reel are displayed on a game grid having a number of symbol positions. During a gaming event, the symbols are randomly arranged via a first spin on game grid to generate a first configuration and a determination is made as to whether the first condition satisfies a predetermined condition. When the predetermined condition is satisfied, the reels or symbols in the symbol positions associated with the predetermined condition are held while the remaining reels or symbol positions are subjected to a second spin to randomly generate a second configuration.



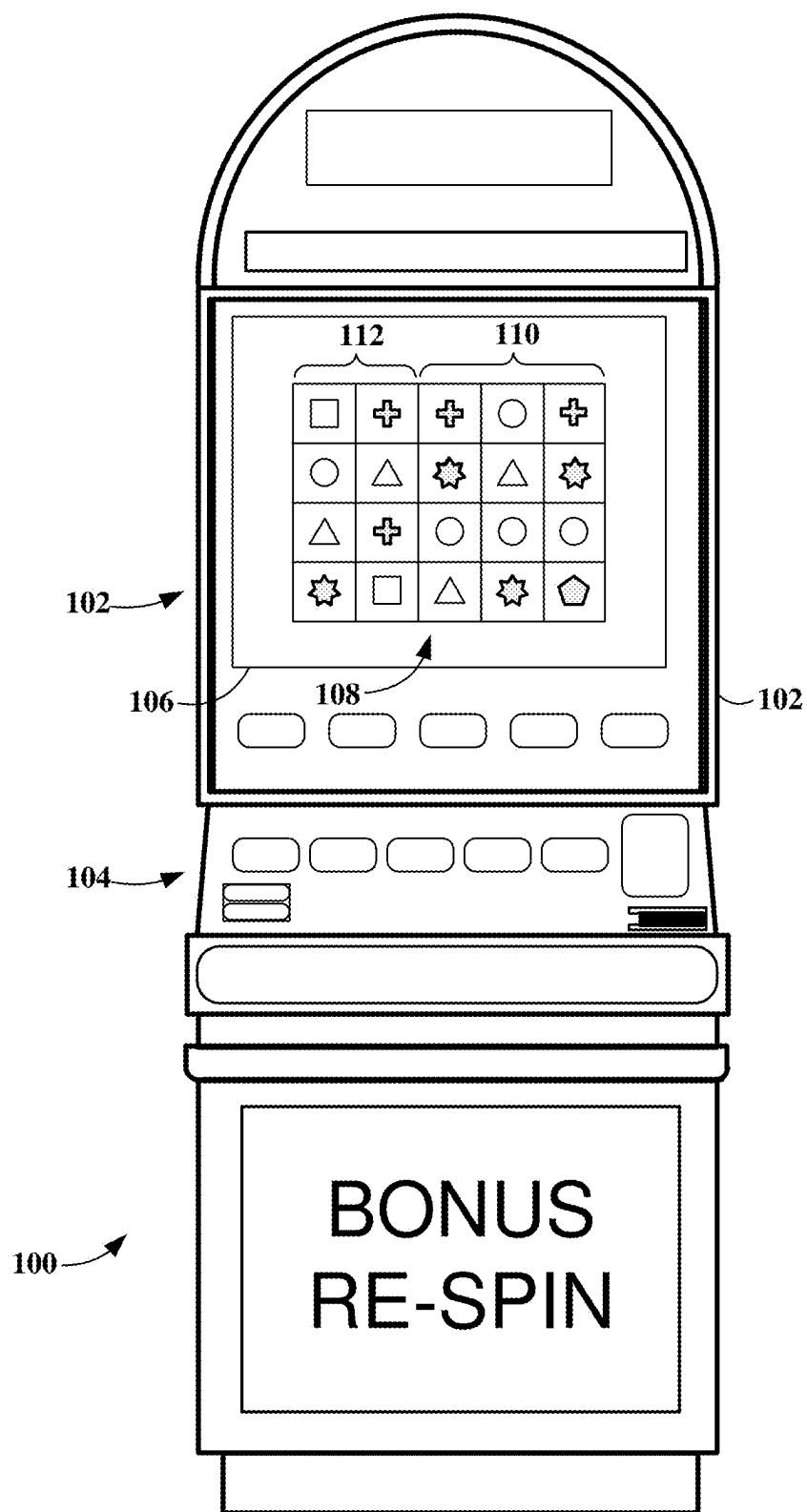


FIG. 1

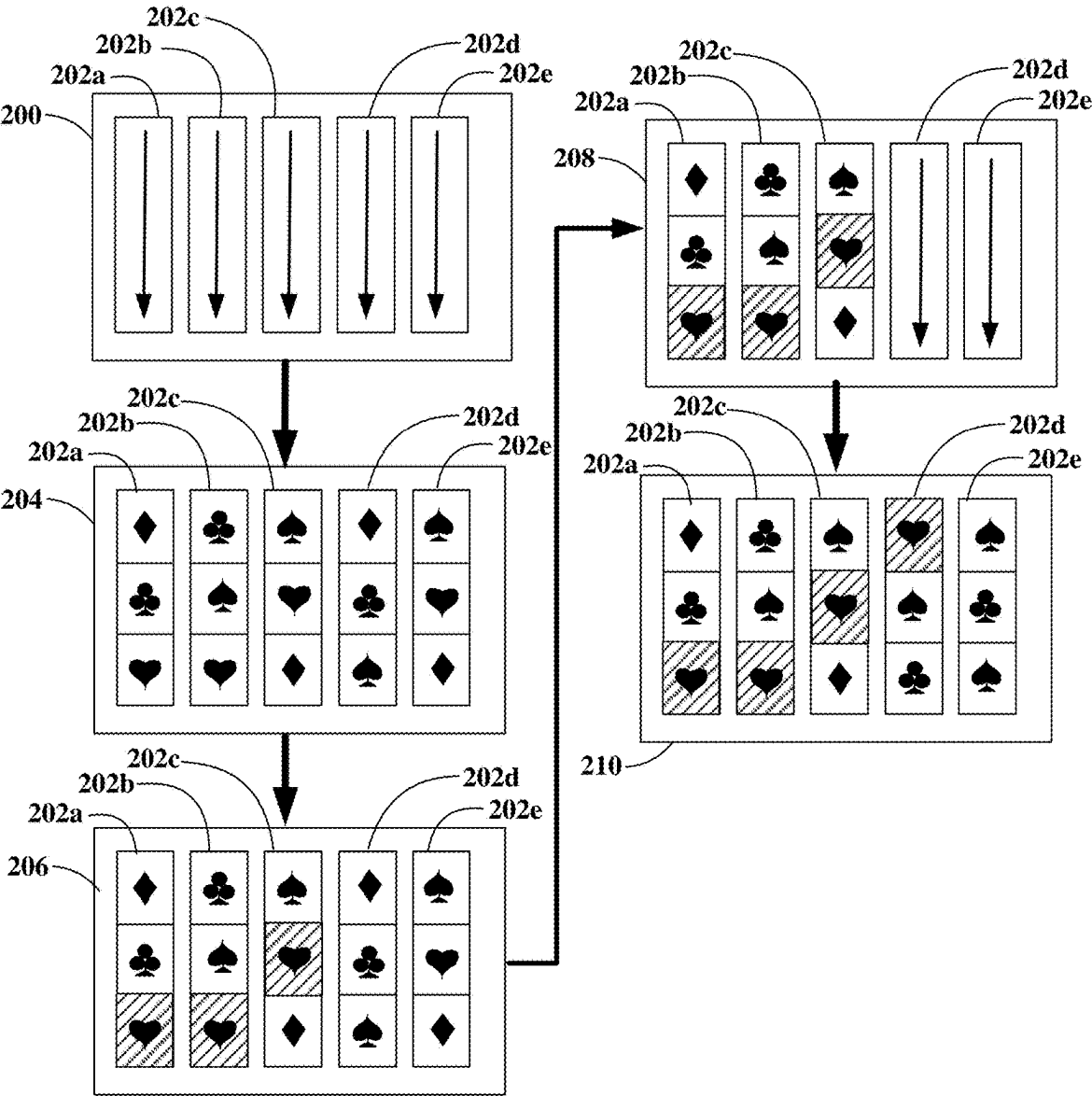


FIG. 2

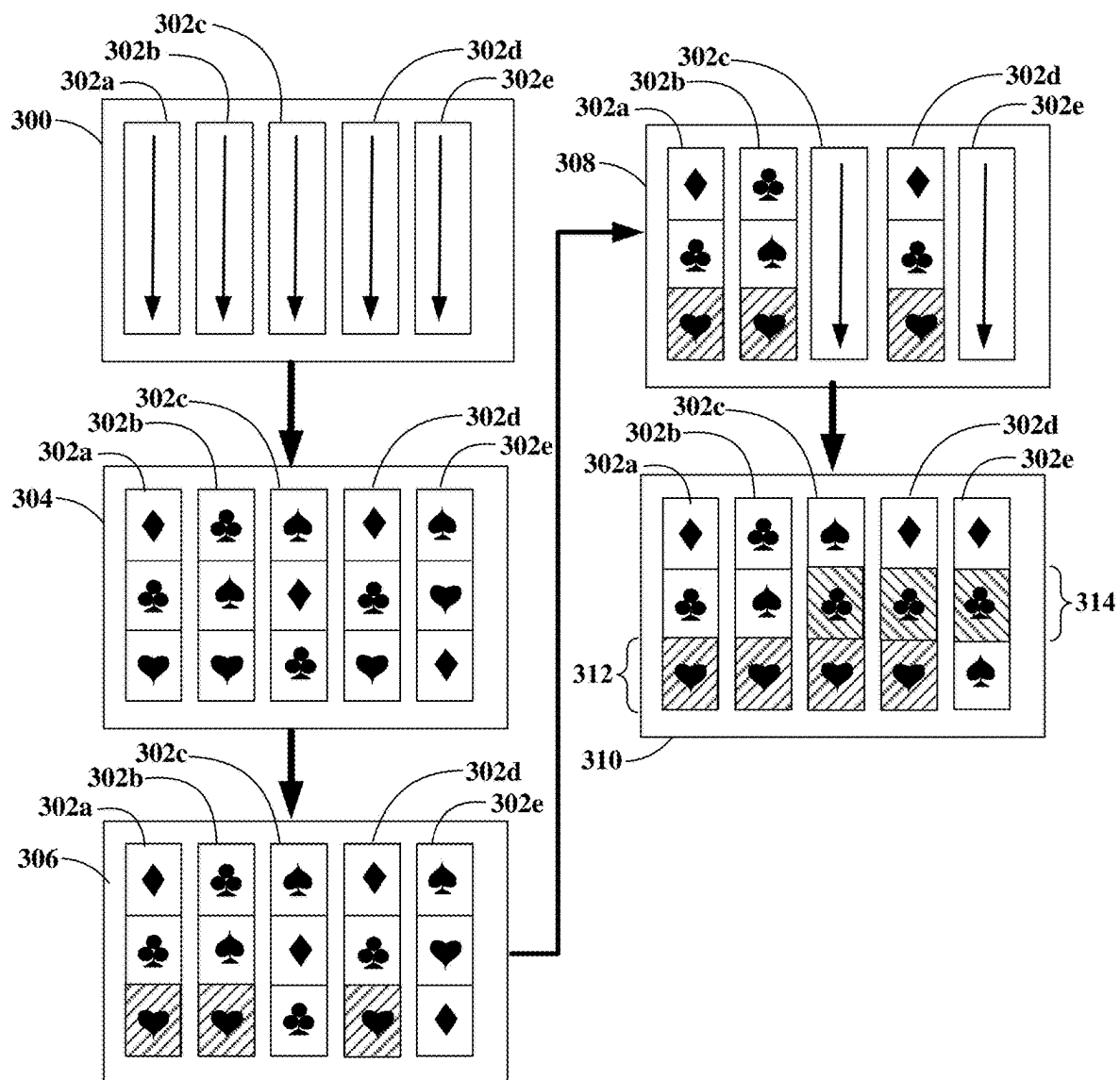


FIG. 3

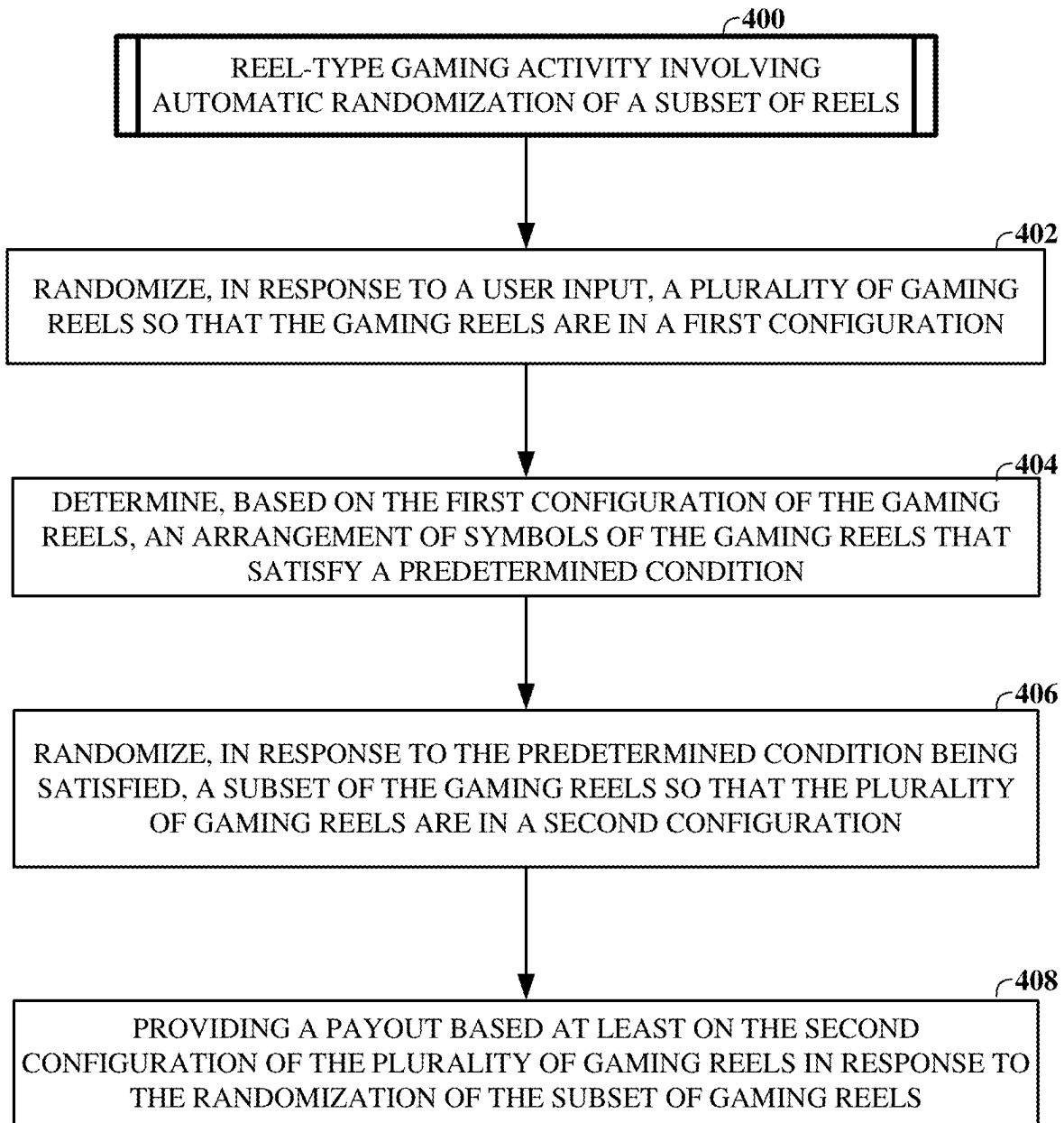


FIG. 4

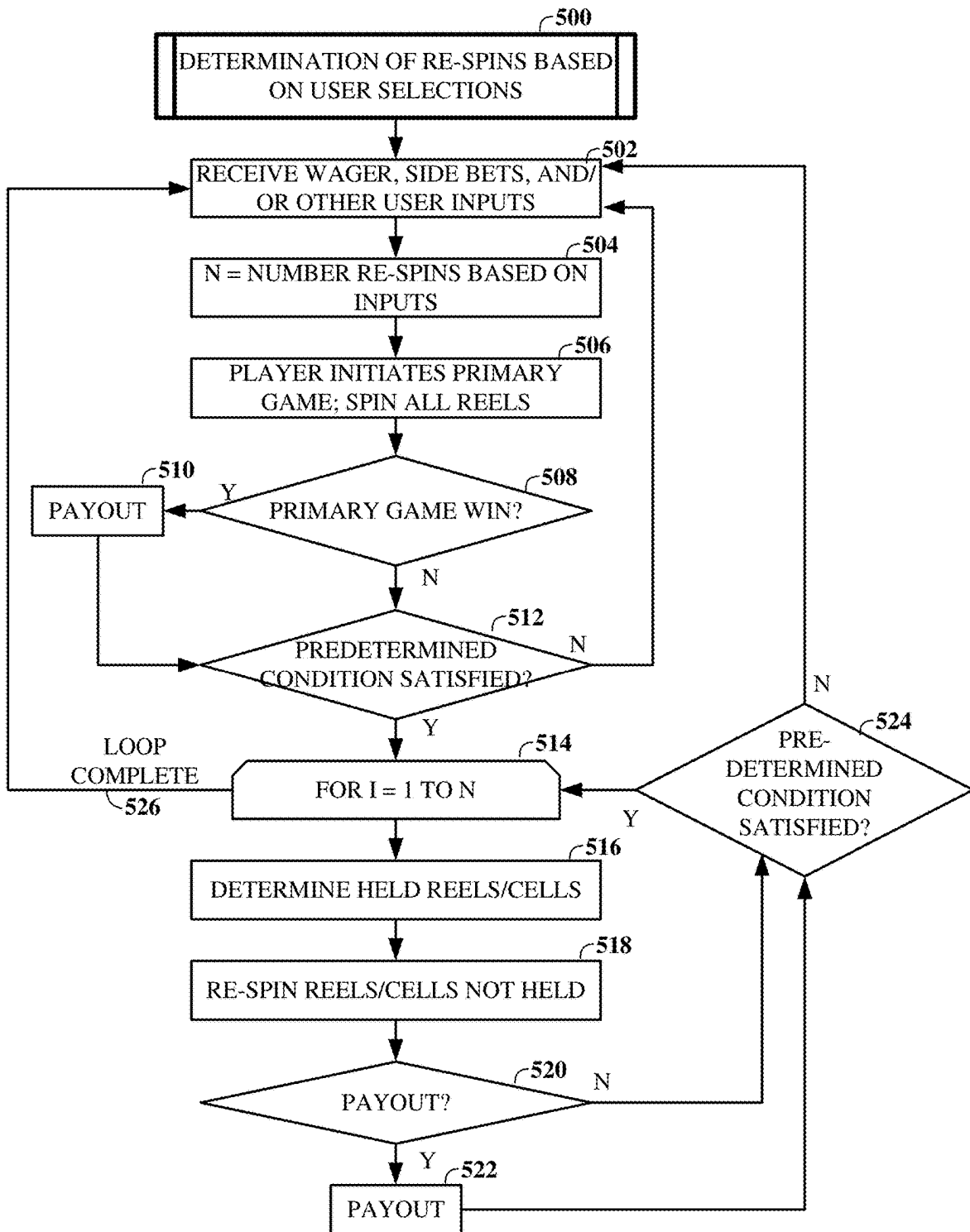


FIG. 5

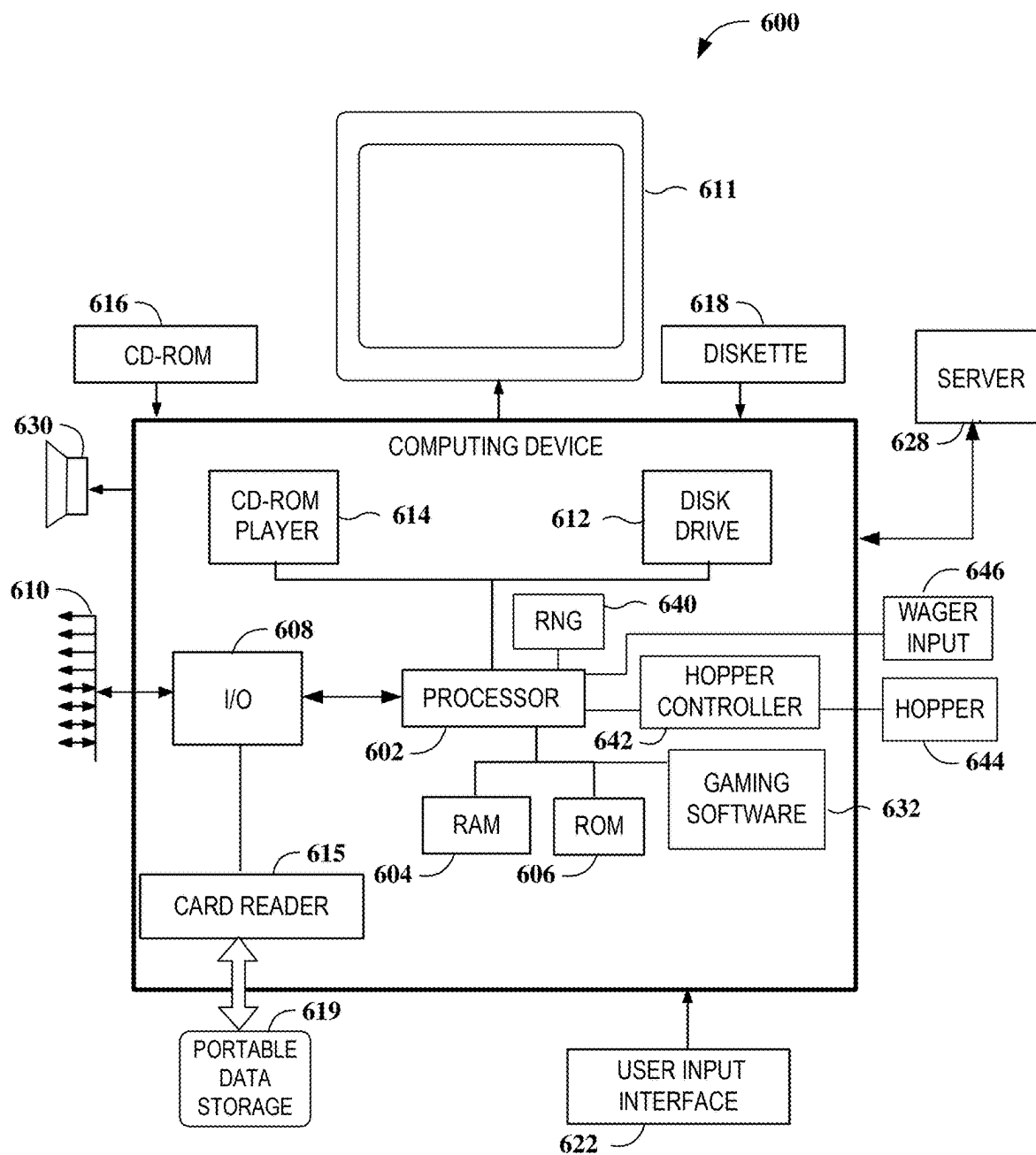


FIG. 6

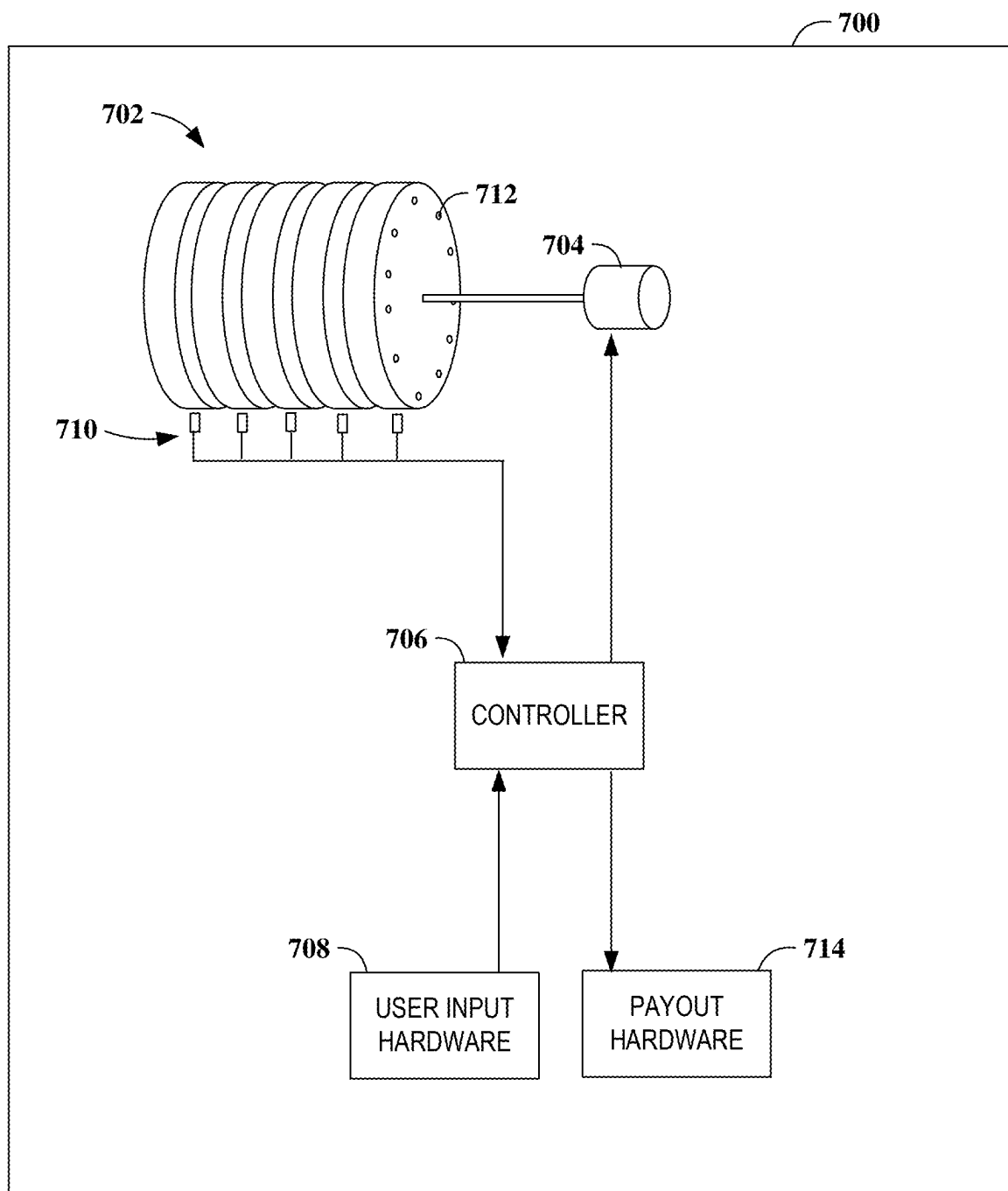
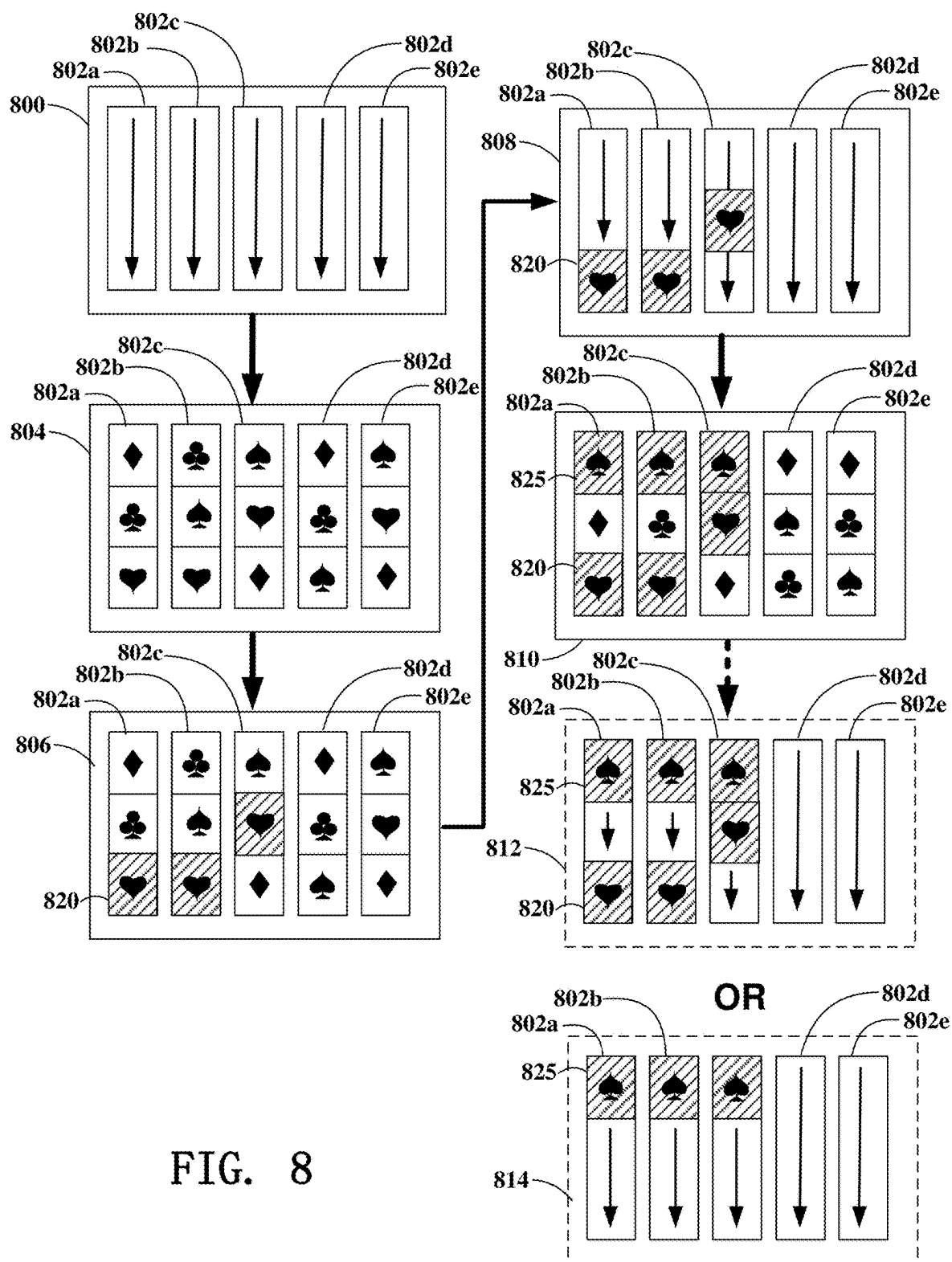


FIG. 7



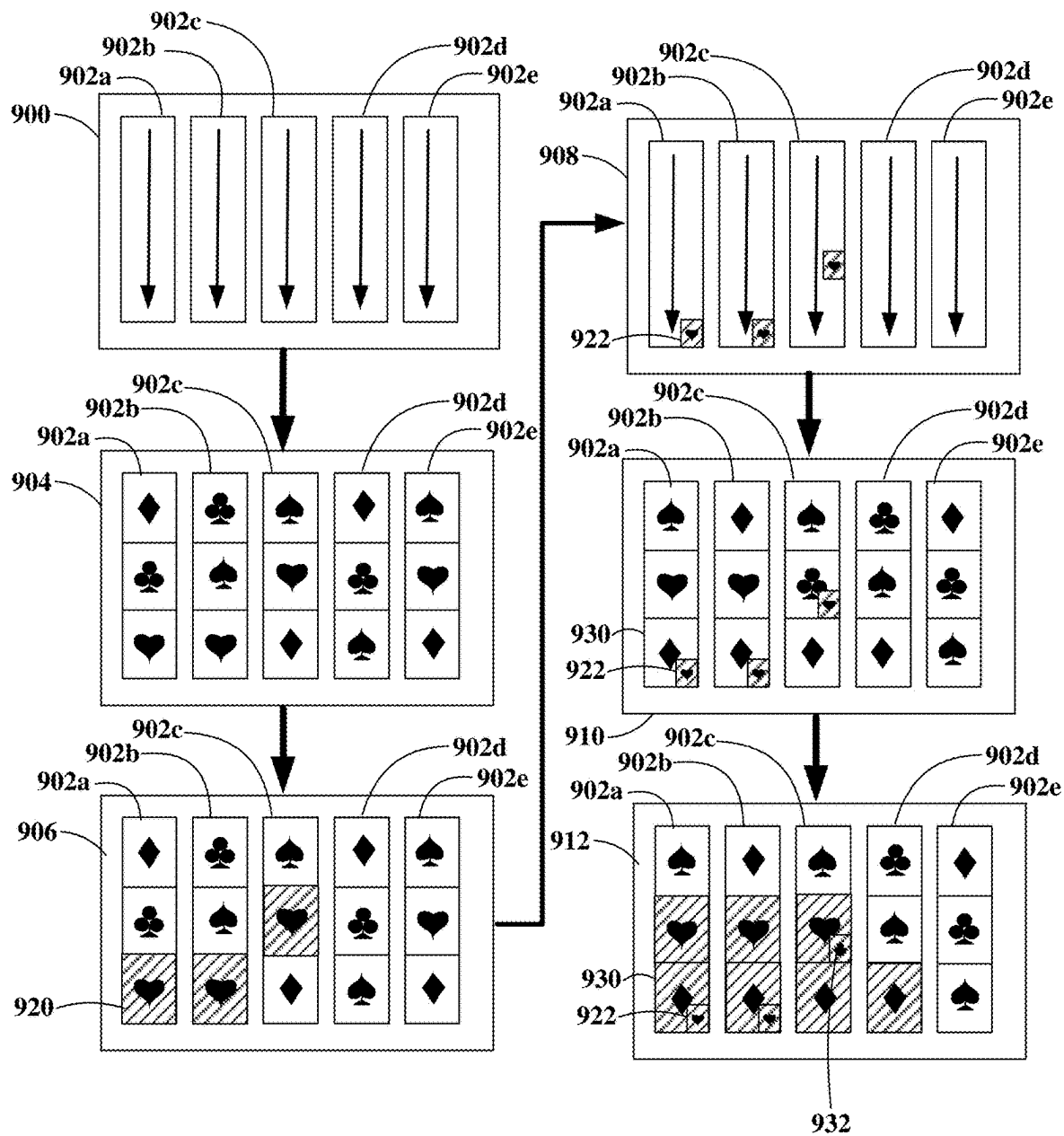


FIG. 9

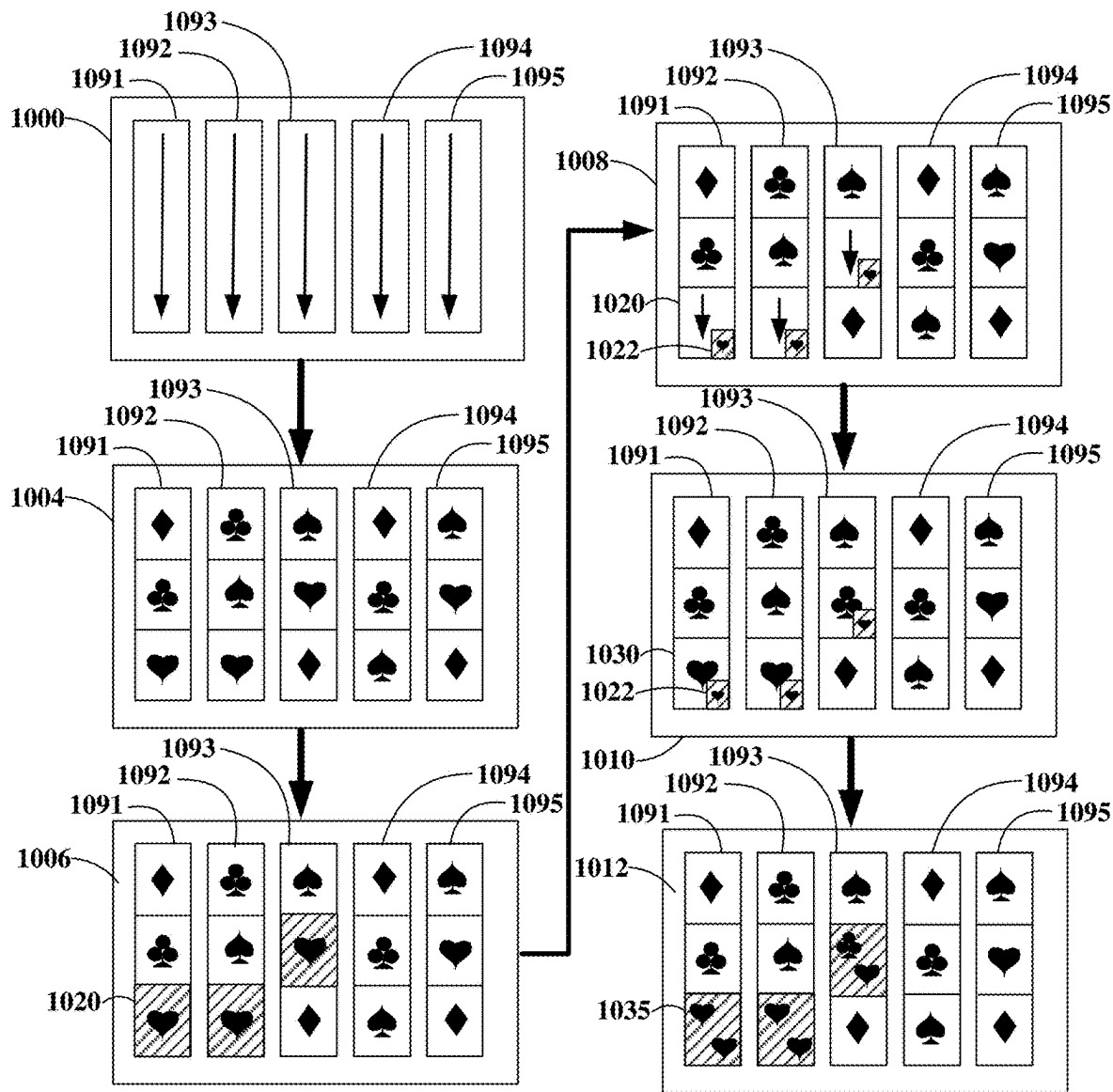


FIG. 10

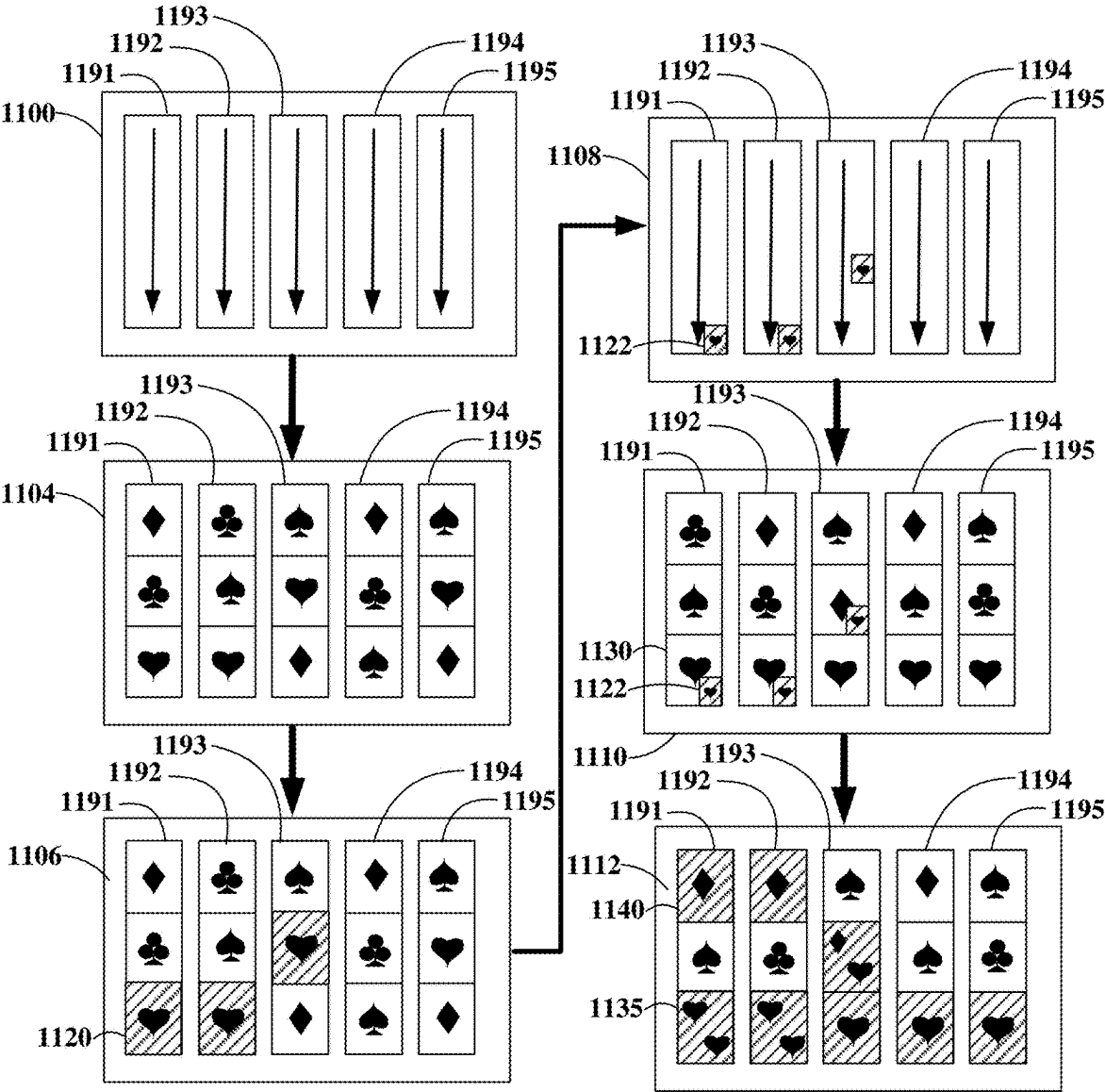


FIG. 11

GAMING DEVICE HAVING MULTIPLE SPINS FOR WINNING OUTCOMES

RELATED APPLICATIONS

[0001] This application is a continuation of U.S. application Ser. No. 18/581,063, filed Feb. 19, 2024, now U.S. Pat. No. 12,307,857, which is a continuation of U.S. application Ser. No. 15/426,040, filed Feb. 6, 2017, now U.S. Pat. No. 11,908,280, which is a continuation of U.S. application Ser. No. 14/923,417, filed on Oct. 26, 2015, now U.S. Pat. No. 9,564,002, which is a continuation of U.S. application Ser. No. 13/975,556, filed on Aug. 26, 2013, now U.S. Pat. No. 9,171,427, which is a continuation-in-part of U.S. application Ser. No. 12/077,719, filed Mar. 20, 2008, now U.S. Pat. No. 8,517,813, which claims the benefit of Provisional Application No. 60/919,362, filed on Mar. 22, 2007, to which priority is claimed pursuant to 35 U.S.C. § 119 (e), all of which are incorporated herein by reference in their entirety.

FIELD OF THE INVENTION

[0002] This disclosure relates generally to games, and more particularly to systems, apparatuses and methods for providing multiple spins for winning outcomes on gaming devices.

BACKGROUND

[0003] Casino games such as poker, slots, and craps have long been enjoyed as a means of entertainment. Almost any game of chance that can be played using traditional apparatus (e.g., cards, dice) can be simulated on a computer. The popularity of casino gambling with wagering continues to increase, as does recreational gambling such as non-wagering computer game gambling. It is also likely that most new games will be implemented, at least in part, using computerized apparatus.

[0004] One reason that casino games are widely implemented on computerized apparatus is that computerized games are highly adaptable, easily configurable and re-configurable, and require minimal supervision to operate. For example, the graphics and sounds included in such games can be easily modified to reflect popular subjects, such as movies and television shows.

[0005] Computer gaming devices can also be easily adapted to provide entirely new games of chance that might be difficult to implement using mechanical or discrete electronic circuits. Because of the ubiquity of computerized gaming machines, players have come to expect the availability of an ever wider selection of new games when visiting casinos and other gaming venues. Playing new games adds to the excitement of “gaming.” As is well known in the art and as used herein, the term “gaming” and “gaming devices” generally involves some form of wagering, and that players make wagers of value, whether actual currency or something else of value, e.g., token or credit. Wagering-type games usually provide rewards based on random chance as opposed to skill. In some jurisdictions, the absence of skill when determining awards during game play is a requirement.

[0006] The present disclosure describes methods, systems, and apparatus that provide for new and interesting gaming experiences, and that provide other advantages over the prior art.

SUMMARY

[0007] To overcome limitations in the prior art described above, and to overcome other limitations that will become apparent upon reading and understanding the present specification, embodiments of the present invention are directed to an apparatus, system, computer readable storage media, and/or method that involve or otherwise facilitate multiple spins during gaming events to provide increased opportunities to receive winning outcomes. Accordingly, a gaming device can be configured to have a plurality of reels with multiple symbols on each reel. A portion of the symbols on each reel are displayed on a game grid having a number of symbol positions. During a gaming event, the symbols are randomly arranged via a first spin on game grid to generate a first configuration and a determination is made as to whether the first condition satisfies a predetermined condition. When the predetermined condition is satisfied, the reels or symbols in the symbol positions associated with the predetermined condition are held while the remaining reels or symbol positions are subjected to a second spin to randomly generate a second configuration.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The invention is described in connection with the embodiments illustrated in the following diagrams.

[0009] FIG. 1 is a diagram of a gaming machine according to an embodiment of the invention.

[0010] FIG. 2 is a sequence diagram showing a sequence of gaming screens in a gaming apparatus according to an embodiment of the invention.

[0011] FIG. 3 is a sequence diagram showing an alternate sequence of gaming screens in a gaming apparatus according to an embodiment of the invention.

[0012] FIG. 4 is a flowchart illustrating a gaming procedure according to an embodiment of the invention.

[0013] FIG. 5 is a flowchart illustrating a procedure for determining re-spins according to an embodiment of the invention.

[0014] FIG. 6 is a block diagram illustrating a computing arrangement according to an embodiment of the invention.

[0015] FIG. 7 is a block diagram illustrating an electro-mechanical apparatus according to an embodiment of the invention.

[0016] FIG. 8 is a sequence diagram showing a sequence of gaming screens in a gaming apparatus according to an embodiment of the invention.

[0017] FIG. 9 is a sequence diagram showing an alternate sequence of gaming screens in a gaming apparatus according to an embodiment of the invention.

[0018] FIG. 10 is a sequence diagram showing another alternate sequence of gaming screens in a gaming apparatus according to an embodiment of the invention.

[0019] FIG. 11 is a sequence diagram showing another alternate sequence of gaming screens in a gaming apparatus according to an embodiment of the invention.

DETAILED DESCRIPTION

[0020] In the following description of various exemplary embodiments, reference is made to the accompanying drawings that form a part hereof, and in which is shown by way of illustration various embodiments in which the invention may be practiced. It is to be understood that other embodi-

ments may be utilized, as structural and operational changes may be made without departing from the scope of the present invention.

[0021] Generally, the present invention relates to a wagering game that may resemble a reel-type gaming apparatus such as slot machines. Generally, a reel-type apparatus provides a gaming activity that involves randomly arranging symbols in such a way as to provide a payout. A commonly implemented form of this apparatus involves using a number of side-by-side circular reels that spin on the same axis. These reels may be mechanical devices (e.g., wheels or hoops) or may be simulated via a computer and video display. The reels have symbols printed on their surface, and the player is provided a monetary award when the selected symbols form a pattern, such as when a line that spans a number of the reels has the same symbol (or satisfies some other pattern or condition). This line that connects symbols is sometimes referred to as the pay line (or payout line), and pay lines may be horizontal, diagonal or other shapes besides a straight line.

[0022] In an apparatus according to an embodiment of the invention, after a reel-spin gaming event, winning combinations are evaluated. If a subset of the reels satisfy a condition, such as forming a winning combination or forming a particular arrangement of symbols, then the subset of the reels remain fixed while the remaining reels are automatically re-spun. The new configuration, which is based on the held subset of reels and newly spun reels, is evaluated to determine any additional winnings. This re-spin of the other reels is automatic, and is not based on a user selection. As such, this feature not does it involve the use of skill on the part of the player.

[0023] In the description that follows, the term “reels,” “reel strips,” and similar mechanically descriptive language may be used to describe various apparatus presentation features. Although the present disclosure may be applicable to both to mechanical and computerized embodiments, and any combination therebetween, the use of mechanically descriptive terms is not meant to be only applicable to mechanical embodiments. Those skilled in the art will understand that, for purposes of providing gaming experiences to players, mechanical elements such as reels may be simulated on a display in order to provide a familiar and satisfying experience that emulates the behavior of mechanical objects. Further, the computerized version may provide the look of a reel (e.g., a linear arrangement of symbols) and inter-reel elements but are randomized in a way different than a spinning reel, such as by randomly and independently changing each cell of the reel that has a symbol. Thus, the term “reels,” “reel strips,” etc. are intended to describe both physical objects and emulation or simulations of those objects using electronic apparatus.

[0024] In various embodiments of the invention, the gaming displays are described in conjunction with the use of data in the form of “symbols.” In the context of this disclosure, a “symbol” refers to a collection of one or more arbitrary indicia or signs that have some conventional significance. In particular, the symbol represents values that can at least be used to determine whether to award a payout. A symbol may include numbers, letters, shapes, pictures, textures, colors, sounds, etc., and any combination therebetween. A win can be determined by comparing the symbol with another symbol. Generally, such comparisons can be performed via software by mapping numbers (or other data structures such

as character strings) to the symbols and performing the comparisons on the numbers/data structures.

[0025] In reference now to FIG. 1, a gaming machine **100** is illustrated that provides a gaming experience according to an embodiment of the invention. The illustrated gaming machine **100** may include a computing system (not shown) to carry out operations according described herein. The gaming machine **100** includes a display **102**, and a user interface **104**, although some or all of the user interface **104** may be provided via the display **102** in touch screen embodiments. The user interface **104** allows the user to control and engage in play of the gaming machine **100**. The particular user interface mechanisms included with user interface **104** may be dependent on the type of gaming machine. For example, the user interface **104** may include one or more buttons, switches, joysticks, levers, pull-down handles, trackballs, voice-activated input, or any other user input system or mechanism that allows the user to play the particular gaming activity.

[0026] The user interface **104** may allow the user to enter coins, bills, or otherwise obtain credits through vouchers, tokens, credit cards, tickets, etc. Various mechanisms for entering such vouchers, tokens, credit cards, coins, tickets, etc. are known in the art. For example, coin/symbol input mechanisms, card readers, credit card readers, smart card readers, punch card readers, radio frequency identifier (RFID) readers, and other mechanisms may be used to enter wagers. It is through the user interface **104** that the user can initiate and engage in gaming activities. While the illustrated embodiment depicts various buttons for the user interface **104**, it should be recognized that a wide variety of user interface options are available for use in connection with the present invention, including pressing buttons, touching a segment of a touch-screen, entering text, entering voice commands, or other known data entry methodology.

[0027] The display device **102** may include one or more of an electronic display, a mechanical display, and fixed display information such as information such as payable information associated with a glass/plastic panel on the gaming machine **100**. The symbols or other indicia associated with the play of the game may be presented on an electronic display device. Generally, the display **102** devotes the largest portion of viewable area to the primary gaming portion **106**. The gaming portion **106** is generally where the visual feedback for any selected game is provided to the user. The gaming portion **106** may render graphical objects such as cards, slot reels, dice, animated characters, and any other gaming visual known in the art. The gaming portion **106** also typically informs players of the outcome of any particular event, including whether the event resulted in a win or loss.

[0028] In the illustrated embodiment, the gaming portion **106** displays a set of primary reels **108**. The reels **108** each include symbols that may be animated so that the symbols appear to be on the surface of a wheel that is rotating vertically when game play is initiated. As is known in the art, when the symbols of the reels stop moving (typically after a random amount of time when physical reel devices are involved), the player may be provided a monetary award if some set of symbols on adjacent reels **108** satisfy some criteria. In addition, some subset of the reels, e.g., subset **110**, may be selected based on this win criteria, such as the row of three matching circles seen in subset **110**. The other reels not in the subset, e.g., subset **112**, are spun again

automatically for an additional turn. The configuration of the both subsets of reels **110**, **112** after the re-spin determines a secondary payout

[0029] The sequence diagram of FIG. 2 shows an example of how a reel-type game that includes automatic, selective re-spins may proceed according to an embodiment of the invention. Screen **200** shows reels **202a-e** being randomized by spinning, as indicated by the vertical arrows. The screen **200** is typically seen after the player has made a wager and initiated play, such as by pulling a lever or pushing a button. The randomization of the symbols associated with the reels **202a-e** may be accomplished in other ways besides using vertically spinning reels, such as by independently randomizing cells within each of the reels **202a-e**. The spinning of the reels **202a-e** seen in screen **200** may be in response to a primary gaming event, a bonus event, or some other play event or feature.

[0030] In screen **204**, the randomization of reels **202a-e** is complete, and wins may be evaluated at this time. The evaluation of wins is shown in screen **206**, which shows shaded cells that correspond to a payline or winning sequence. The indicated win involves a subset of the reels **202a-e**, in particular reels **202a-c**. A sequence of symbols formed by reels **202a-c** matches a predetermined criteria (e.g., three neighboring cells having a high symbol) and therefore reels **202d-e** are re-spun, as is shown in screen **208**. Screen **210** shows an additional win evaluation that occurs after reels **202d-e** have stopped spinning. As can be seen by the additional highlighted cell in reel **202d**, an additional match has occurred, and additional payout is provided based on this match.

[0031] The re-spinning of some of the reels may occur only based on certain conditions occurring with the other reels. For example, the re-spin may only occur if the matching of the other reels involves high symbols, wild symbols, or other special symbols. The re-spin may be activated by the player placing a side bet before the initial spin, and this could be allowed instead of or in addition to special symbols that trigger a re-spin. In some cases, special symbols or side bets may be able to activate more than one re-spin. In the example of FIG. 2, if the player had paid to activate three re-spins, if needed, then reels **202d-e** would be re-spun and wins re-evaluated two more times similar to what is shown in screens **208**, **210**.

[0032] Another example of how multiple re-spins may be applied may be demonstrated by referring again to screen **210** of FIG. 2. This screen **210** resulted from a re-spin, and as described above, triggered an additional winning event due to the matching symbol appearing in reel **202d** with symbols of reels **202a-c**. If the player has paid for more than one re-spin in such a case, the re-spin may be applied only to the remaining reel **202e**, instead of to both **202d** and **202e**. Thus, if the player has wagered a side bet for three additional spins, reel **202e** may be re-spun two more times in an attempt to get yet another win, e.g., a sequence of five matching symbols. Even where the side bet only activates a single re-spin (or where no side bet is required), the game may automatically spin additional reels (e.g., reel **202e**) if a reel that was re-spun once (e.g., reel **202d**) results in an additional winning event.

[0033] In another variation, instead of re-spinning reels **202e-f**, only the symbols that are next in sequence on the pay line for which a pay could be made would spun. For example, if the paylines are limited to horizontal lines, and

the initial win included three symbols along the bottom row of a first set of reels, only the symbols of the remaining reels along the bottom row will be re-spun. This latter example may be implemented in reel type games where individual cells can be randomized independent of other cells on the same reel.

[0034] It will be appreciated that a re-spin may be automatically awarded even when the first spin did not result in a win. One example of this according to an embodiment of the invention is shown in FIG. 3. Screen **300** shows reels **302a-e** being randomized by spinning, as indicated by the vertical arrows. In screen **304**, the randomization of reels **302a-e** is complete, and in this case, the reels **302a-e** may not form a winning combination. However, as is indicated by shaded cells in screen **306**, three matching symbols lie on a payline, although not in sequences. If the re-spin feature is activated and can be triggered by such an event, the reels **302a**, **302b**, and **302d** are held, and reels **302c** and **302e** are re-spun, as is shown in screen **308**.

[0035] Screen **310** shows the win evaluation that occurs after reels **302c** and **302e** have stopped spinning. As can be seen by the shaded row **312**, the re-spinning has cause a four-in-a-row match, and a payout is provided based on this match. As is also seen in screen **310**, a three-in-a-row match has also occurred as indicated in row **314**. This latter sequence **314** may or may not be included in the payout. This could depend on the rules of the particular game, or could be dependent on particular wagers.

[0036] In reference now to FIG. 4, an example procedure **400** is illustrated for providing a gaming experience according to embodiments of the invention. In response to a user input, a plurality of gaming reels is randomized **402** so that the gaming reels are in a first configuration. Based on the first configuration of the gaming reels, it is determined **404** whether an arrangement of symbols of the gaming reels satisfy a predetermined condition. In response to the predetermined condition being satisfied, a subset of the gaming reels is randomized **406** so that the plurality of gaming reels are in a second configuration. The subset of gaming reels is randomized **406** independently of additional user inputs occurring after the determination of the predetermined condition. The gaming reels not in the subset of reels are held while the subset of gaming reels is randomized. A payout is provided **408** based at least on the second configuration of the plurality of gaming reels in response to the randomization of the subset of gaming reels.

[0037] As discussed hereinabove, the triggering of selected reel re-spins may be conditioned on a user action taken before the full set of reels is activated. This user action may be a side wager or other selection, and may include the ability to trigger more than one re-spin. In reference now to FIG. 5, a procedure **500** illustrates how user inputs before game play is initiated may affect re-spins according to an embodiment of the invention. A player will provide input **502** such as wagers, side bets, etc., that may or may not enable the re-spinning of selected reels. Based on this input, **502**, a variable **N** is determined **504** based on the wager. In some variations, **N** may be a constant (e.g., set to one) or may be a function of the wager amount.

[0038] The player initiates game play **506** and all reels are spun or otherwise randomized. A determination **508** is made if this spin results in a payout, in which case the payout **510** may be provided or otherwise indicated to the player. In either event, a test **512** is made for the existence of a

predetermined condition, typically based on arrangement of the reels. In some embodiments, a primary game win **508** may be part of the condition, so that if determination **508** is no, then determination **512** is also always no. In other arrangements, even a non-winning arrangement may still satisfy the condition, e.g., the showing of special symbols in reels and/or other predetermined patterns.

[0039] If the predetermined condition **512** is satisfied, then a loop **514** is entered, and may be re-entered multiple times depending on the value of **N** previously determined **540** as well as other conditions. The loop **514** involves determining **516** which reels are to be held and which are to be re-spun **518**. Note that in some game embodiments, individual cells may be randomized instead of re-randomizing a whole reel. After the re-spinning **518**, a payout is determined **520** and payout **522** may be provided.

[0040] In some variations, the loop **514** may test **524** for the existence of the predetermined condition before continuing. For example, additional re-spins may be dependent on previous re-spins resulting in additional wins **522**. If the condition **524** is not satisfied, or the loop is complete, as indicated by path **526**, the procedure resumes its initial input condition **502**. Note that if **N=0**, (e.g., player made no side wager), then the loop **514** may terminate **526** even if the predetermined condition **512** is satisfied.

[0041] As may now be readily understood, one or more devices may be programmed to play various embodiments of the invention. The present invention may be implemented as a casino gaming machine such as a slot machine or other special purpose gaming kiosk as described hereinabove, or may be implemented via computing systems operating under the direction of local gaming software, and/or remotely-provided software such as provided by an application service provider (ASP). The casino gaming machines utilize computing systems to control and manage the gaming activity. An example of a representative computing system capable of carrying out operations in accordance with the invention is illustrated in FIG. 6.

[0042] Hardware, firmware, software or a combination thereof may be used to perform the various gaming functions, display presentations and operations described herein. The functional modules used in connection with the invention may reside in a gaming machine as described, or may alternatively reside on a stand-alone or networked computer. The computing structure **600** of FIG. 6 is an example computing structure that can be used in connection with such electronic gaming machines, computers, or other computer-implemented devices to carry out operations of the present invention.

[0043] The example computing arrangement **600** suitable for performing the gaming functions in accordance with the present invention typically includes a central processor (CPU) **602** coupled to random access memory (RAM) **604** and some variation of read-only memory (ROM) **606**. The ROM **606** may also represent other types of storage media to store programs, such as programmable ROM (PROM), erasable PROM (EPROM), etc. The processor **602** may communicate with other internal and external components through input/output (I/O) circuitry **608** and bussing **610**, to provide control signals, communication signals, and the like.

[0044] The computing arrangement **600** may also include one or more data storage devices, including hard and floppy disk drives **612**, CD-ROM drives **614**, card reader **615**, and other hardware capable of reading and/or storing informa-

tion such as DVD, etc. In one embodiment, software for carrying out the operations in accordance with the present invention may be stored and distributed on a CD-ROM **616**, diskette **618**, access card **619**, or other form of media capable of portably storing information. These storage media may be inserted into, and read by, devices such as the CD-ROM drive **614**, the disk drive **612**, card reader **615**, etc. The software may also be transmitted to the computing arrangement **600** via data signals, such as being downloaded electronically via a network, such as the Internet. Further, as previously described, the software for carrying out the functions associated with the present invention may alternatively be stored in internal memory/storage of the computing device **600**, such as in the ROM **606**.

[0045] The computing arrangement **600** is coupled to the display **611**, which represents a display on which the gaming activities in accordance with the invention are presented. The display **611** represents the "presentation" of the video information in accordance with the invention, and may be any type of known display or presentation screen, such as LCD displays, plasma display, cathode ray tubes (CRT), digital light processing (DLP), liquid crystal on silicon (LCOS), etc. Where the computing device **600** represents a stand-alone or networked computer, the display **611** may represent a standard computer terminal or display capable of displaying multiple windows, frames, etc. Where the computing device is embedded within an electronic gaming machine, the display **611** corresponds to the display screen of the gaming machine/kiosk. A user input interface **622** such as a mouse, keyboard/keypad, microphone, touch pad, trackball, joystick, touch screen, voice-recognition system, etc. may be provided. The display **611** may also act as a user input device, e.g., where the display **611** is a touchscreen device.

[0046] Chance-based gaming systems such as slot machines, in which the present invention is applicable, are governed by random numbers and processors, as facilitated by a random number generator (RNG). In particular, the fixed and dynamic symbols generated as part of a gaming activity may be produced using one or more RNGs. RNGs are known in the art, and may be implemented using hardware, software operable in connection with the processor **602**, or some combination of hardware and software. The present invention is operable using any known RNG, and may be integrally programmed as part of the processor **602** operation, or alternatively may be a separate RNG controller **640**.

[0047] The computing arrangement **600** may be connected to other computing devices or gaming machines, such as via a network. The computing arrangement **600** may be connected to a network server **628** in an intranet or local network configuration. The computer may further be part of a larger network configuration as in a global area network (GAN) such as the Internet. In such a case, the computer may have access to one or more web servers via the Internet.

[0048] Other components directed to gaming machine implementations include manners of gaming participant payment, and gaming machine payout. For example, a gaming machine including the computing arrangement **600** may also include a hopper controller **642** to determine the amount of payout to be provided to the participant. The hopper controller may be integrally implemented with the processor **602**, or alternatively as a separate hopper controller **642**. A hopper **644** may also be provided in gaming

machine embodiments, where the hopper serves as the mechanism holding the coins/tokens of the machine. The wager input module 646 represents any mechanism for accepting coins, tokens, coupons, bills, electronic fund transfer (EFT), tickets, credit cards, smart cards, membership cards, etc., for which a participant inputs a wager amount. It will be appreciated that the primary gaming software 632 may be able to control payouts via the hopper 644 and controller 642 for independently determined payout events.

[0049] Among other functions, the computing arrangement 600 provides an interactive experience to players via input interface 622 and output devices, such as the display 611, speaker 630, etc. These experiences are generally controlled by gaming software 632 that controls a primary gaming activity of the computing arrangement 600. The gaming software 632 may be temporarily loaded into RAM 604, and may be stored locally using any combination of ROM 606, drives 612, or media player 614. The primary gaming software 632 may also be accessed remotely, such as via the server 628 or the Internet.

[0050] The primary gaming software 632 in the computing arrangement 600 according to embodiments of the present invention provides a floating reel-type gaming experience as defined hereinabove. For example, the software 632 may present, by way of the display 611, a plurality of gaming reels each having a plurality of symbols that are randomly arranged in response to gaming events. The software 632 controls the reels by randomizing, in response to a user input, the reels so that the gaming reels are in a first configuration. Based on the first configuration of the gaming reels, the software 632 determines an arrangement of the symbols that satisfy a predetermined condition. In response to the predetermined condition being satisfied, the software 632 randomizes a subset of the gaming reels so that the plurality of gaming reels are in a second configuration. The software 632 causes payout devices 642, 644 to provide a payout based at least on the second configuration of the plurality of gaming reels in response to the randomization of the subset of gaming reels.

[0051] It will be appreciated that the above functionality described in relation to a computer implemented gaming apparatus may also be applied to electromechanical apparatus as well. In reference now to FIG. 7, an apparatus 700 according to an embodiment of the invention is illustrated. The apparatus includes mechanical reels 702 that generally have symbols printed on an outer surface. The reels 702 are controlled by one or more motors 704, which receive commands from a controller 706. The motor 704 may be rotary or linear (e.g., solenoid and/or linear stator device). The motor 704 may cause the reels 702 to turn, or a mechanical device such as a lever (not shown) may cause the reels to turn in response to user activation. In the latter case, the motor 704 may cause the reels 702 to randomly stop, such as by applying a braking force to a hub or shaft.

[0052] The controller 706 may include digital and/or analog circuitry that implements the logic and control functions of the apparatus. The controller 706 receives user inputs via input hardware 708, and in response may cause the motor 704 to randomize all of the reels 702 into a first configuration. The controller may be coupled to sensors 710 that detect reel positions, such as by optical, magnetic, or other markers located on the reels 702, as illustrated by marker 712. Based on the first configuration, the controller

706 may cause a payout to be provided by way of payout hardware 714. Additionally, the controller determines an arrangement of the reels 702 that satisfies a predetermined condition. In response to the predetermined condition being satisfied, the controller 706 randomizes a subset of the reels 702 so that the reels 702 are in a second configuration. The controller 706 causes payout hardware 714 to provide a payout based at least on the second configuration of the reels 702 in response to the randomization of the subset of reels.

[0053] FIG. 8 is a sequence diagram showing a sequence of gaming screens in a gaming apparatus according to an embodiment of the invention. Referring to FIG. 8, gaming display or screen 800 shows reels 802a-e being randomized by spinning the reels, as indicated by the vertical arrows. The screen 800 is typically seen after the player has made a wager and initiated play, such as by pulling a lever or pushing a button. The randomization of the symbols associated with the reels 802a-e may be accomplished in other ways besides using vertically spinning reels, such as by independently randomizing cells within each of the reels 802a-e. The spinning of the reels 802a-e seen in screen 800 may be in response to a primary gaming event, a bonus event, or some other play event or feature.

[0054] In screen 804, the randomization of reels 802a-e is complete, and wins may be evaluated at this time. The evaluation of wins is shown in screen 806, which shows shaded symbol positions or cells 820 that correspond to a payline or winning sequence. The indicated win involves a subset of the symbol positions on the game grid reels 802a-e, in particular symbol positions 820. A sequence of symbols formed by symbol positions 820 matches a predetermined criteria (e.g., three neighboring symbol positions having a high symbol) and therefore the symbols at the winning symbol positions 820 are held or have their position maintained while reels 802a-e are re-spun, as is shown in screen 808. In some embodiments, the held symbol positions 820 are shown above the spinning reels (802a-c in this instance). Here, symbols from the game reels that would have typically landed in the held symbol positions 820 can be hidden and replaced by the symbols at the held symbol positions, or these symbols can be displaced upward (effectively nudging the reels back a symbol) or downward. In alternate embodiments, where each symbol position is associated with an independent reel, the reels associated with the held symbol positions 820 may not be spun or activated.

[0055] Screen 810 shows an additional win evaluation that occurs after reels 802a-e have stopped spinning. As can be seen by the additional highlighted symbol positions 825 on the game grid, an additional match has occurred, and an additional payout is provided based on this match. In some embodiments, the award for the first winning combination at the held symbol positions 820 is repaid again, as well. That is, all displayed winning combinations are paid with each game outcome. In other embodiments, only newly formed winning symbol combinations generate awards for the player.

[0056] The re-spinning of the reels 802a-e may occur only based on certain conditions occurring after an initial spin. For example, the re-spin may only occur if the winning condition on the reels 802a-e involves high symbols, wild symbols, or other special symbols. The re-spin may be activated by the player placing a side bet before the initial spin, and this could be allowed instead of or in addition to special symbols that trigger a re-spin. In some cases, special

symbols or side bets may be able to activate more than one re-spin. These re-spins may continue as long as a predefined condition is met. For example, re-spins may occur as long as a previous spin has resulted in a new winning symbol combination, or a new winning symbol combination with an award value higher than award values associated with previously received (and held) symbol combinations. In the example of FIG. 8, optional screens **812** and **814** show two different embodiments of re-spin variations.

[0057] In the embodiment shown in screen **812**, the three spade symbols associated with symbol positions **825** and held along with the three heart symbols in the originally held symbol positions **820** while the reels **802a-e** are re-spun. Alternatively, as shown in screen **814**, only the latest symbol positions **825** associated with a winning condition may be held while the reels **802a-e** are re-spun. As previously mentioned, re-spins may only be triggered if a new symbol combination is associated with a larger award than the awards associated with previously held symbol combinations. If this condition was configured in the gaming device for the screens **812** and **814**, then the three spade symbols in symbol positions **825** would have to be associated with a larger award than the three heart symbols in symbol positions **820** for the re-spin to occur.

[0058] Another example of how multiple re-spins may be applied may be demonstrated by referring again to screen **810** of FIG. 8. In this example, the player must pre-pay for the ability to re-spin the reels **802a-e**. Screen **810** resulted from a re-spin, and as described above, triggered an additional winning event due to the matching symbol appearing in symbol positions **825**. If the player has paid for more than one re-spin in such a case, the re-spin may be applied only to the remaining symbol positions not in either winning combination as shown in screen **812**. Alternatively, only symbol positions related to a latest or highest paying symbol combination may be held as shown in screen **814**. Thus, if the player has wagered a side bet for three additional spins, the game reels **802a-e** may be re-spun two more time in an attempt to get yet another win. Even where the side bet only activates a single re-spin, the game may automatically spin the reels **802a-e** if another winning condition has been satisfied as described above.

[0059] In another variation, instead of re-spinning reels **802a-e**, only the symbol positions that are next in sequence on the pay line associated with a symbol positions **820** associated with a winning condition for which a pay could be made would spun. For example, if the paylines are limited to horizontal lines, and the initial win included three symbols along the bottom row of a first set of reels, only the symbol positions of the remaining reels along the bottom row will be re-spun. This latter example may be implemented in reel type games where individual cells can be randomized independent of other cells on the same reel.

[0060] FIGS. 9-11 include additional embodiments that are shown in sequences of gaming screens in a gaming apparatus. Any of these embodiments can be modified using the variations discussed above. However, a further description of these variations with these embodiments is not included so as not to obscure the features of these different embodiments. Referring to FIG. 9, a game sequence is shown for embodiments where symbols associated a winning combination as a result of a first spin are marked or otherwise associated with their initial symbol position while the game reels are re-spun. Here, new symbols are received

in the symbol positions of the held or marked symbols, providing for a variety of symbol evaluation options. In particular, game reels **902a-e** are spun in screen **900** during a first gaming event, which results in a first game outcome shown in screen **904** when the reels come to rest. This first game outcome is evaluated for a winning condition (or other triggering event that generates a re-spinning of the reels **902a-e**) in screen **906**. Here, the symbol positions **920** are shaded or otherwise indicated to show a winning symbol combination.

[0061] As shown in screen **908**, the symbols associated with this winning condition are marked or otherwise associated with their winning symbol positions **920** as markings **922** while the game reels **902a-e** are re-spun. In some embodiments, the markings **922** may include shrinking the symbol image to a portion of the symbol position **920** and spinning the reels **902a-e**. In other embodiments, the markings **922** may be configured in other ways to associate the symbol images with the corresponding symbol positions **920** while the reels **902a-e** are spinning. As shown in screen **910**, the game reels **902a-e** have come to a rest to display a second game outcome. Here, new symbols **930** have been positioned in the previous symbol positions **920** that were associated with the winning condition in the first game outcome.

[0062] The second outcome is then evaluated for winning conditions. Here, the new symbols **930** or the markings **922** of the previous symbols used in the winning condition of the first game outcome may be used in evaluating the second game outcome. In some embodiments, symbol combinations using either the new symbols **930** or symbols in the markings **922** are determined. That is each permutation or combination of symbols from the new symbols **930** and marking **922** is considered and evaluated. In some embodiments, an optimal combination is determined for one of the new symbols **930** or corresponding marking **922** to appear as the symbol in the associated symbol position. For example, as shown in screen **912**, the diamond symbols in the new symbols **930** have been maintained in the two lower symbol positions on reels **902a** and **902b** since a four diamond payline can be created on a horizontal payline on the bottom row. However, the middle symbol position in the third reel **902c** has switched symbols so that the previous marking **922** (the heart) becomes the symbol for final evaluation while the previous new symbol **930** (the club) becomes a new marking **932**, so that a three heart symbol combination along a horizontal middle payline can be formed. Here, only the final symbols and not the markings (after the determination if the symbols and markings should be interchanged) are used in the final evaluation of the game reels **902a-e** to determine symbol combinations associated with awards and/or winning conditions that may trigger additional re-spins.

[0063] In other embodiments, the determination of the whether to use the new symbols **930** or the markings **922** in symbol combination evaluations for awards is done on a payline-by-payline basis. Thus, in a final evaluation the symbol shown in the a symbol position having a new symbol **930** and a marking **922** may alternate between the new symbol and the symbol associated with the marking **922** depending on which payline result is being shown. This results in an optimal payout result for each played payline.

[0064] Additional variations may also be possible, such as implementing rules as to when markings can be inter-

changed with symbols. For example, one rule may not allow all of the previous markings from replacing the new symbols at the corresponding symbol positions. In another example, a player may select which of the two symbols to use between the new symbol and the marking. In other embodiments (shown in FIGS. 10 and 11), the new symbol may be combined with the marking symbol to form a multi-indicia symbol that includes properties of both symbols. As described above, multiple re-spins may be performed in some embodiments when triggering conditions for the one or more additional re-spins is met.

[0065] In FIGS. 10 and 11 embodiments are shown in the form of a game progression where symbols meeting a winning condition (or other triggering condition) are held and associated with a corresponding symbol position while new symbols are generated for the symbol positions (FIG. 10) or the game reels are re-spun (FIG. 11) to generate new symbols on a game grid. The previously held symbols are then merged with the new symbols to form multi-indicia symbols which can be used in a variety of ways to provide awards.

[0066] Referring to FIG. 10, a game screen 1000 shows reels 1091-1095 being spun to generate a first game outcome, which is shown on screen 1004. This first game outcome is evaluated for a winning condition (or other triggering event that generates a re-spinning of the reels 1091-1095) in screen 1006. Here, the symbol positions 1020 are shaded or otherwise indicated to show a winning symbol combination. As shown in screen 1008, the symbols associated with this winning condition are marked or otherwise associated with their winning symbol positions 1020 as markings 1022 while the game reels 1091-1093 are re-spun for the winning symbol positions 1020. In some embodiments, individual reels are spun for the symbol positions 1020, while in other embodiments, new symbols are randomly selected for the symbol positions 1020 associated with the winning condition without spinning any reels. Regardless of which embodiment is implemented, new symbols 1030 are shown in the symbol positions 1020 associated with the winning condition from the first game outcome, as shown in screen 1010. These new symbols are then merged or combined with the symbols of the markings 1022 (the symbols used in the winning condition in the first game outcome) to form multi-indicia symbols 1035, as shown in screen 1012.

[0067] These multi-indicia symbols 1035 may be evaluated in a variety of manners. In one embodiment, each symbol of the multi-indicia symbol 1035 is evaluated as a separate symbol. Hence, for screen 1012, a five symbol heart pay may be awarded since each of the hearts in the first two multi-indicia symbols 1035 on reels 1091 and 1092 would be identified or counted as two heart symbols, and the heart from the multi-indicia symbol appearing in the middle location of reel 1093 would be identified or counted as a single heart. Although these symbols on the multi-indicia symbols 1035 may be shown to be individually identified and awarded a similar award to five single heart symbols, a game processor may simply recognize them as special symbol that is associated with a different entry in a coded payable stored in the memory of the gaming device.

[0068] In other embodiments, the multi-indicia symbol may act as a multiplier, such as doubling the value of the three heart pay. Thus, if this embodiment was implemented in screen 1012, a player may be award a pay value of 4 times

the value of a normal 3-symbol heart pay. This is because each of the double heart multi-indicia symbols would be considered a “2x” multiplier. Multiple multipliers received on a payline may be summed together or multiplied together. Alternatively, a multiplier may be randomly assigned to each multi-indicia symbol 1035 having a double symbol, where in some embodiments, only the highest multiplier would be used to modify a value of a winning symbol combination pay. Other variations in evaluation techniques are possible and considered within the scope of this concept.

[0069] As mentioned above, in some embodiments, subsequent re-spins may occur when predefined conditions are met. In the embodiment shown in FIG. 10, both symbols of the multi-symbol indicia 1035 may be shrunk or otherwise marked to the associated symbol location, and a new symbol may be generated in the symbol location by spinning the reels or otherwise randomly choosing another symbol. In these cases, a subsequent re-spin may generate 3-symbol multi-indicia symbols, or 4 or more multi-indicia symbols after subsequent re-spins. Note that in this embodiment, other symbols in the symbol positions are maintained when the symbols in the symbol locations associated with a winning condition are re-spun. Although not shown in FIG. 10, other embodiments, may cascade or otherwise shift the reels 1091-1095 or symbols of the reels down to replace the removed (and marked) symbols used in the winning condition of the first game outcome.

[0070] The embodiment shown in the game progression of FIG. 11 is similar to the one shown in FIG. 10, except the entire reels re-spin for a second game outcome instead of just the symbol positions used in a winning condition. In particular, a game screen 1100 shows reels 1191-1195 being spun to generate a first game outcome, which is shown on screen 1104. This first game outcome is evaluated for a winning condition (or other triggering event that generates a re-spinning of the reels 1191-1195) in screen 1106. Here, the symbol positions 1120 are shaded or otherwise indicated to show a winning symbol combination.

[0071] As shown in screen 1108, the symbols associated with this winning condition are marked or otherwise associated with their winning symbol positions 1120 as markings 1122 while the game reels 1191-1195 are re-spun. New symbols 1130 are shown in the symbol positions 1120 associated with the winning condition from the first game outcome, as shown in screen 1110. These new symbols are then merged or combined with the symbols of the markings 1122 (the symbols used in the winning condition in the first game outcome) to form multi-indicia symbols 1135, as shown in screen 1112. These multi-indicia symbols 1135 may be evaluated in a similar manner as described above. Thus, for screen 1112, a player may receive a 7-heart pay (bottom horizontal payline), a 5-heart pay (payline used in first game outcome), and a 3-diamond pay. Alternatively, if multi-indicia symbols 1135 can only be used once, the player would only receive a 7-heart pay and a 3-diamond pay.

[0072] The foregoing description of the exemplary embodiments has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. For example, the present invention is equally applicable in electronic or mechanical gaming machines, and is also applicable to live table versions of

gaming activities that are capable of being played in a table version (e.g., machines involving poker or card games that could be played via table games).

[0073] Some embodiments of the invention have been described above, and in addition, some specific details are shown for purposes of illustrating the inventive principles. However, numerous other arrangements may be devised in accordance with the inventive principles of this patent disclosure. Further, well known processes have not been described in detail in order not to obscure the invention. Thus, while the invention is described in conjunction with the specific embodiments illustrated in the drawings, it is not limited to these embodiments or drawings. Rather, the invention is intended to cover alternatives, modifications, and equivalents that come within the scope and spirit of the inventive principles set out in the appended claims.

1. A method of operating a gaming apparatus for participating in games of chance that includes a display device having a game grid of symbol positions, a memory storing data related to a plurality of game reels each having a plurality of symbols that includes at least one special symbol wherein each of the plurality of game reels is associated with one of the symbol positions in the game grid, a user interface including at least one user input to enable initiation of a game of chance which includes randomizing at least some of the plurality of game reels to present symbols in the symbol positions of the game grid, a wager input device structured to identify and validate player assets and to permit play the games of chance when the player assets are provided, and a processor, the method comprising:

- spinning each of the plurality of game reels;
- identifying a bonus event based on at least a plurality of the special symbols presented in respective ones of the symbol positions in the game grid;
- holding the special symbols in their respective symbol positions such that game reels associated with special symbols do not change during the bonus event;
- re-spinning each of the game reels not associated with the special symbols to provide one of the plurality of symbols, which may or may not be one of the special symbols, in the symbols positions of the game grid;
- holding newly presented ones of the special symbols, if any, in their respective symbol positions, and re-spin each of the game reels not associated with the special symbols, as long as a predefined condition is met; and
- calculating a final payout based on at least the special symbols that were held during the spin and re-spins of the plurality of game reels.

2. The method of claim 1, wherein at least one symbol other than the special symbols is a color.

3. The method of claim 1, wherein the special symbols comprise numbers.

4. The method of claim 1, wherein the method includes providing the final payout after the predefined condition has not been met.

5. The method of claim 1, wherein the method includes providing the final payout after at least one repeat of the held special symbols and random presentation of symbols in each of the remaining plurality of symbol positions that does not hold the special symbols.

6. The method of claim 1, wherein the method includes re-spinning each of the game reels not associated with the special symbols N times.

7. A method of operating a gaming apparatus for participating in games of chance that includes a display device having a game grid of symbol positions where the symbol positions organized into a plurality of columns and a plurality of rows within the game grid, a memory storing data related to a plurality of game reels each having a plurality of symbols that includes at least one special symbol, a user interface including at least one user input to enable initiation of a game of chance which includes randomizing at least some of the plurality of game reels to present symbols in the symbol positions of the game grid, a wager input device structured to identify and validate player assets and to permit play the games of chance when the player assets are provided, and a processor, the method comprising:

- spinning each of the plurality of game reels;
- identifying a bonus event based on at least a plurality of the special symbols presented in respective ones of the symbol positions in the game grid;
- holding the special symbols in their respective symbol positions such that the held special symbols do not change during the bonus event;
- for each of the symbol positions not associated with the special symbols, randomly determining whether to display a special symbol in that respective symbol position;
- holding newly presented ones of the special symbols, if any, in their respective symbol positions, and randomly re-determining whether to display a special symbol in the symbols positions not currently associated with the special symbols, as long as a predefined condition is met; and
- calculating a final payout based on at least the special symbols that were held during the bonus event.

8. The method of claim 7, wherein the method includes randomly re-determining whether to display a special symbol in the symbols positions not associated with the special symbols N times.

9. The method of claim 7, wherein the predefined condition is met when a new symbol position is randomly determined to display a special symbol.

10. The method of claim 7, wherein at least one symbol other than the special symbols is a color.

11. The method of claim 7, wherein the special symbols comprise numbers.

12. The method of claim 7, wherein the method includes providing the final payout after the predefined condition has not been met.

13. The method of claim 7, wherein the method includes providing the final payout after at least one repeat of the held special symbols and random presentation of symbols in each of the remaining plurality of symbol positions that does not hold the special symbols.

14. A method of operating a slot game apparatus for participating in a slot game event, the slot game apparatus including a display device configured to display a game grid including a plurality of columns of reel cells and a plurality of rows of reel cells where each of the plurality of the reels cells is independently capable of presenting one of a plurality of symbols therein, a user interface including at least one user input to enable initiation of the slot game events which includes randomizing and presenting the symbols in their respective reel cells, a wager input device structured to identify and validate player assets and to permit play the slot

game events when the player assets are provided, and a processor, the method comprising:

- (a) randomly presenting a slot game event on the display device in response to a signal received from the user interface;
- (b) identifying a bonus-initiating event based on a plurality of special symbols being presented on the game grid during the slot game event;
- (c) holding the special symbols at their respective ones of the plurality of reel cells;
- (d) randomly determining for each of the remaining reel cells that does not currently display the special symbols whether to present a special symbol in that respective reel cell;
- (e) holding any newly presented ones of the special symbols at their respective ones of the remaining plurality of reel cells;
- (f) continuing (d) and (e) until a condition has been satisfied; and
- (g) providing a final payout based on the special symbols when the condition has been satisfied.

15. The method of claim **14**, wherein the method includes providing the final payout by calculating a first winning

payout amount based on the special symbols presented when the bonus-initiating event is identified, and providing the final payout by adding the first winning payout amount to any other payouts based on the special symbols held in response to continuing (d) and (e).

16. The method of claim **14**, wherein the method includes continuing (d) and (e) until the condition has been satisfied by continuing (d) and (e) until at least a second random presentation of one of the plurality of symbols in the remaining plurality of reel cells that does not hold the special symbols occurs.

17. The method of claim **14**, wherein the method includes continuing (d) and (e) until the condition has been satisfied by continuing (d) and (e) until none of the remaining reel cells are determined to present a special symbol in (d).

18. The method of claim **14**, wherein the method includes continuing (d) and (e) until the condition has been satisfied by performing (d) and (e) at least N times.

19. The method of claim **14**, wherein the special symbols include numbers.

20. The method of claim **14**, wherein at least one symbol other than the special symbols is a color.

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